



Industrial USB TO TTL Converter FT232RL

The **SmartElex Industrial USB to TTL Converter (FT232RL)** is a reliable **USB-to-UART (TTL)** interface module designed for seamless communication between computers and TTL-based devices such as microcontrollers, sensors and embedded systems. Using the industry-proven FT232RL chipset, it offers stable data transfer, wide OS compatibility, and selectable 5V/3.3V logic levels, making it ideal for development, debugging, and industrial applications.

Specifications

- **Model:** USB to UART TTL (FT232RL)
- **USB Connector:** USB Type-A
- **Power Supply:** 5V (from USB)
- **UART Logic Levels:** 5V / 3.3V (jumper selectable)
- **Baud Rate:** 300 bps to 3 Mbps
- **Interface:** USB ↔ UART (TTL)
- **Operating Systems:** Windows 7/8/8.1/10/11, Linux, macOS, Android, WinCE
- **Signals Supported:** TXD, RXD, RTS, CTS, DTR, DSR, DCD, RI, CBUS

Features

- Based on **FT232RL** – stable and industry-standard USB-UART bridge
- Supports **high-speed communication up to 3 Mbps**
- **5V / 3.3V VCCIO output** for powering external TTL devices
- Wide **OS driver support** (plug-and-play on most systems)
- Full **hardware flow control** support (RTS/CTS)
- Ideal for **programming, debugging, logging and testing**
- Suitable for **industrial and development environments**



USB to TTL Converter (FT232) –

Pin Name	Function	Input / Output	Notes / Connection
VCC	Power output from USB (logic supply for module)	—	Typically 5V (some boards provide 3.3V via regulator)
5V	Direct 5V from USB	Output	Can power external device (limited current)
GND	Ground reference	—	Common ground with MCU/device
TXD	Serial data transmit	Output	Connect to MCU RX
RXD	Serial data receive	Input	Connect to MCU TX
RTS	Request To Send (hardware flow control)	Output	Connect to MCU CTS (optional)
CTS	Clear To Send (hardware flow control)	Input	Connect to MCU RTS (optional)

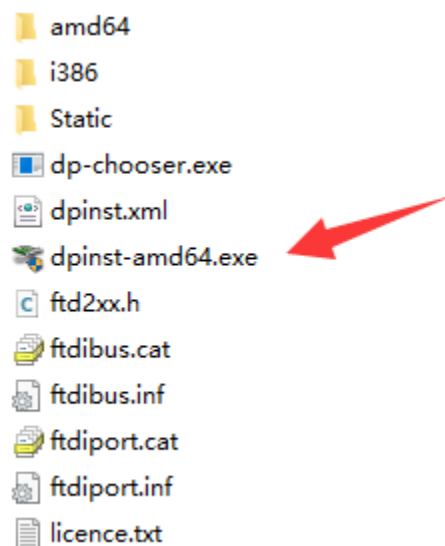
How to Use

Windows –

There are two drivers for FT232: VCP and D2XX.

- VCP (Virtual COM Port): The PC side will be recognized as a serial port, using the serial protocol.
- D2XX driver: The PC side will be recognized as a USB port and needs to be operated using API functions.
- This product provides a VCP example, so we will only discuss the installation of the VCP driver below.
- Download the driver: [FT232_Driver_64](#)

The following is an example of the driver installation steps for the Windows 10 64-bit system:





Welcome to the Device Driver Installation Wizard!

This wizard helps you install the software drivers that some computers devices need in order to work.

To continue, click Next.

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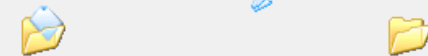
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The drivers are now installing...



Please wait while the drivers install. This may take some time to complete.

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Completing the Device Driver Installation Wizard

The device driver installation wizard did not update any of your software for your hardware devices because it was not better than the software you currently have installed.

Driver Name	Status
✓ FTDI CDM Driver Packa...	Ready to use
✓ FTDI CDM Driver Packa...	Ready to use

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Finish

Cancel

Device Connection (Windows)

1. Install the **FT232 VCP driver**.
 2. Connect the **FT232 USB UART Board** to the PC USB port.
 3. Windows will automatically detect the device.
 4. Right-click **Start Menu** → **Device Manager**.
 5. Open **Ports (COM & LPT)** →
You will see **USB Serial Port (COMx)**.
-

COM Port Setting (Optional – Windows)

If you want to change the COM number:

1. Open **Device Manager**
2. Right-click **USB Serial Port (COMx)** → **Properties**
3. Go to **Port Settings** → **Advanced**
4. Select a **free COM port number** (not used by other devices)
5. Click **OK** → **Apply**

Useful when software requires a fixed COM port.

Device Connection (Linux – Ubuntu / Raspberry Pi)

Most Linux systems have **built-in FT232 drivers**.

Steps:

```
lsusb  
ls /dev/ttyUSB*
```

Example output:

```
/dev/ttyUSB0
```

Open serial using **minicom**:

```
sudo minicom -D /dev/ttyUSB0
```

Works on **Ubuntu, Raspbian, CentOS**, etc.
No manual driver installation required in most cases.